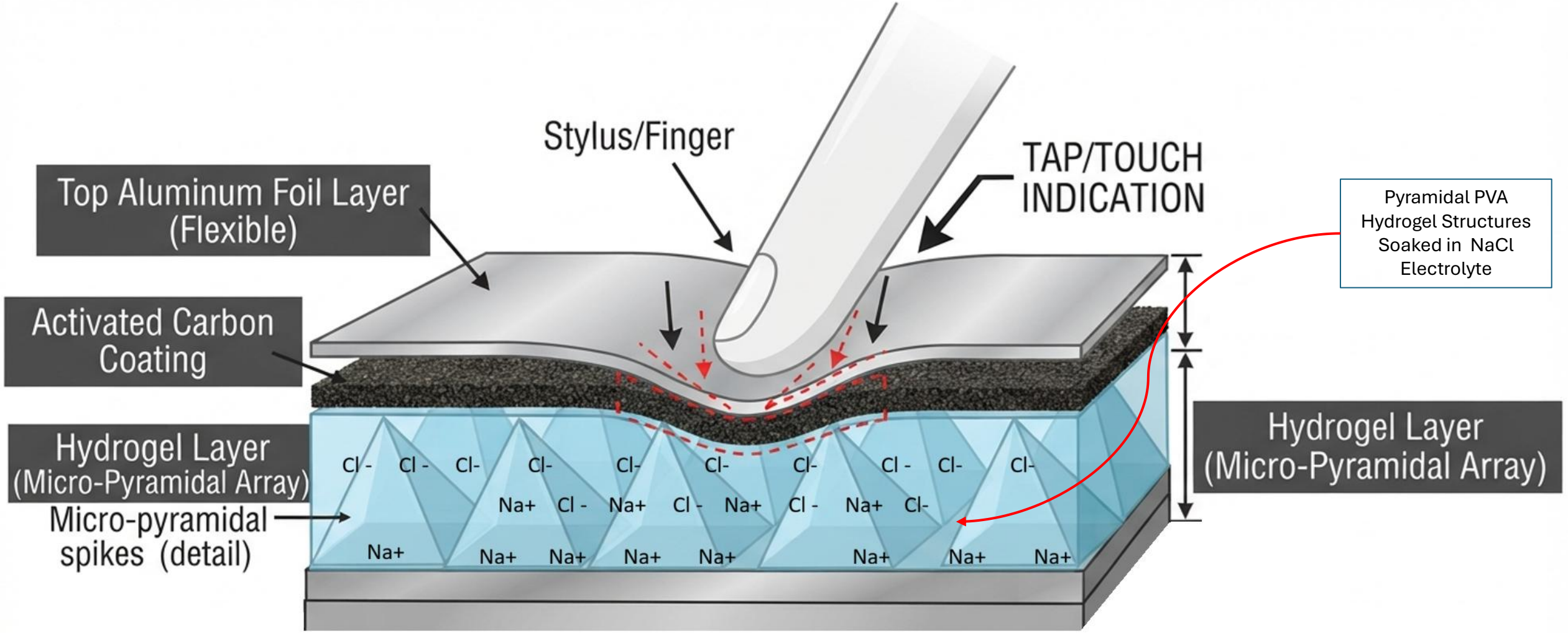
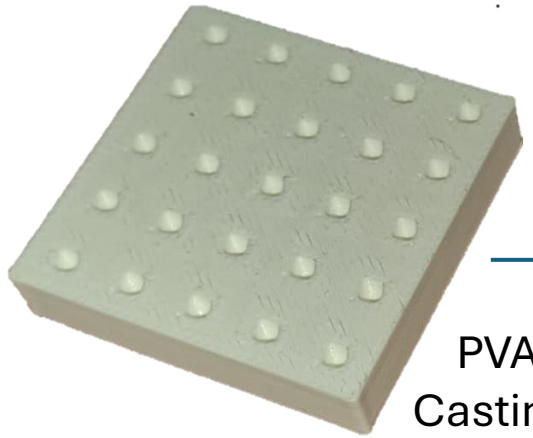
Mechanical Stimuli Based Touch and Tap Sensor to render resistive Output

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Converts touch into Resistance change

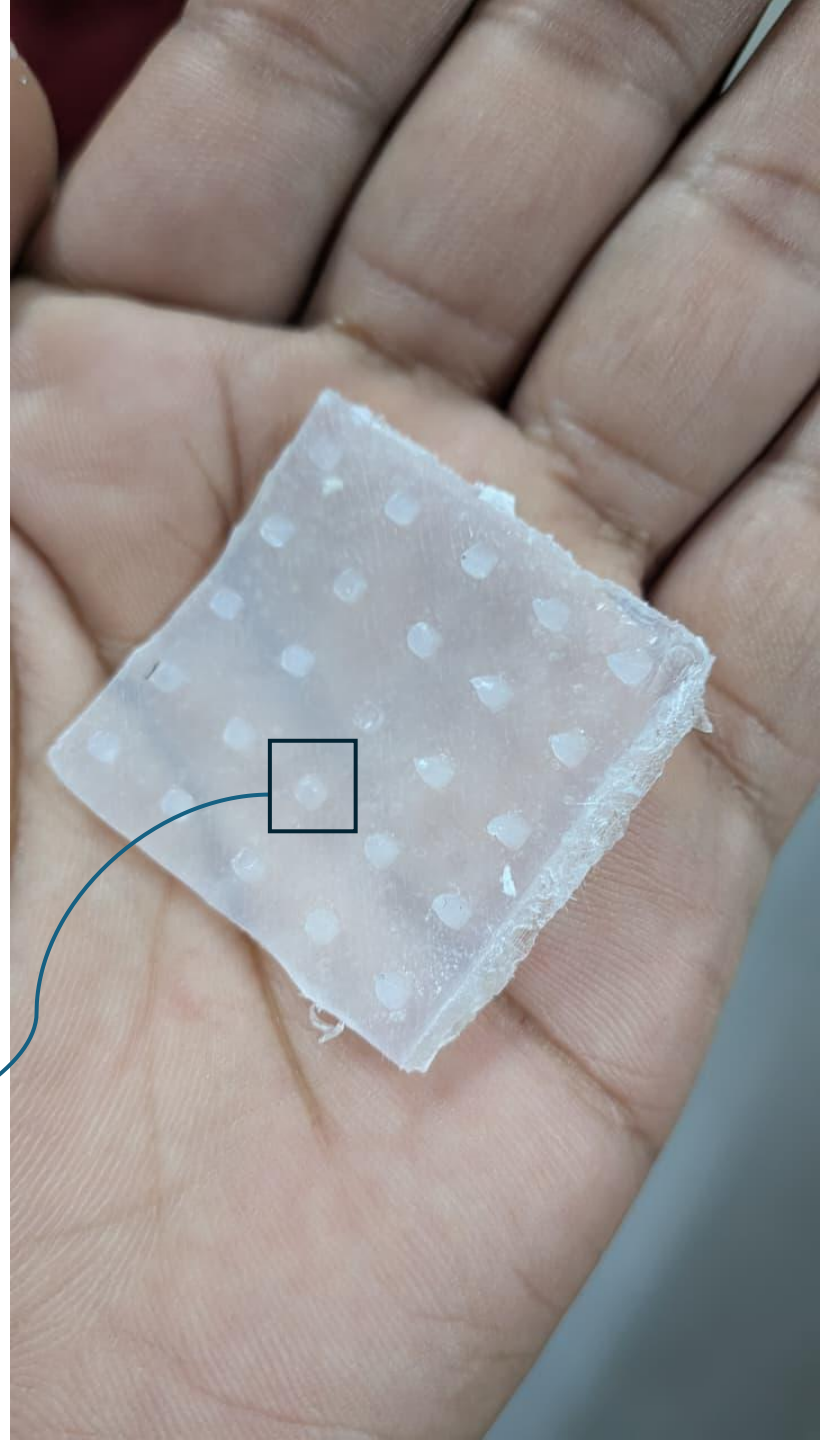
3D Printed PLA Template



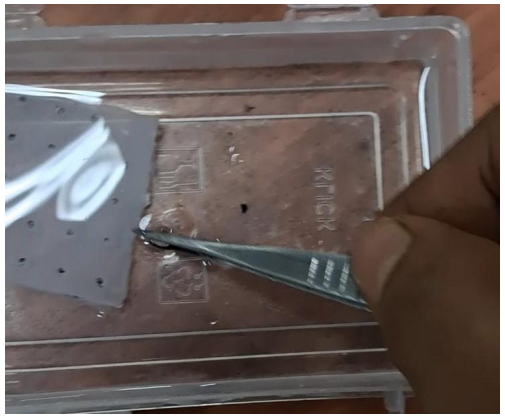
PVA Hydrogel Casting, 13% PVA



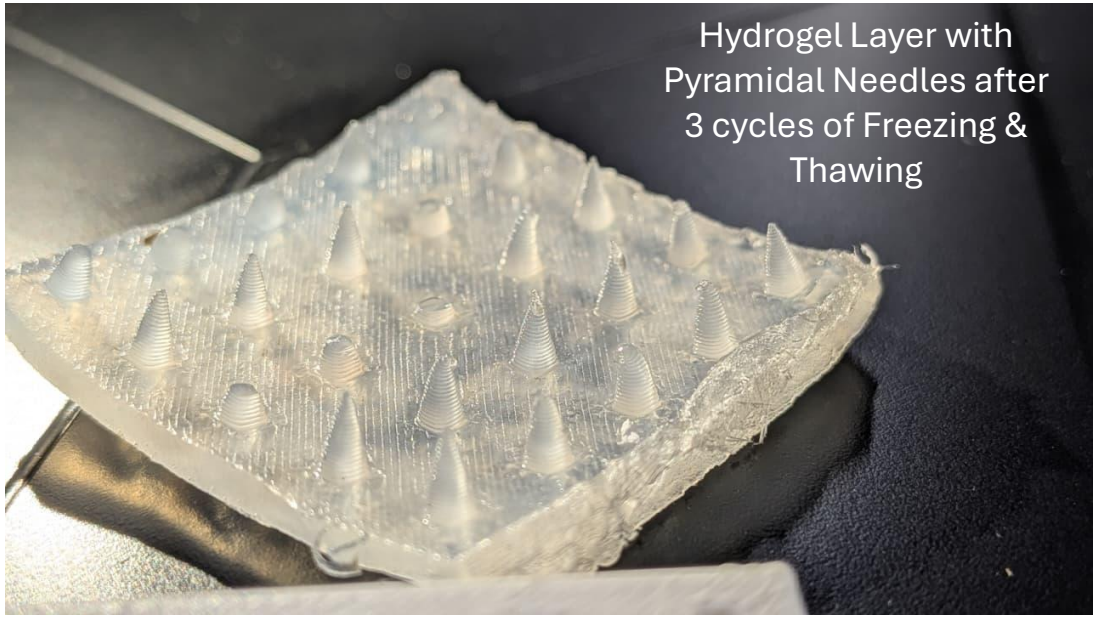
3 Cycles of Freezing & Thawing, (20 hr at -20 degree C) and 4 hr at Room temperature.



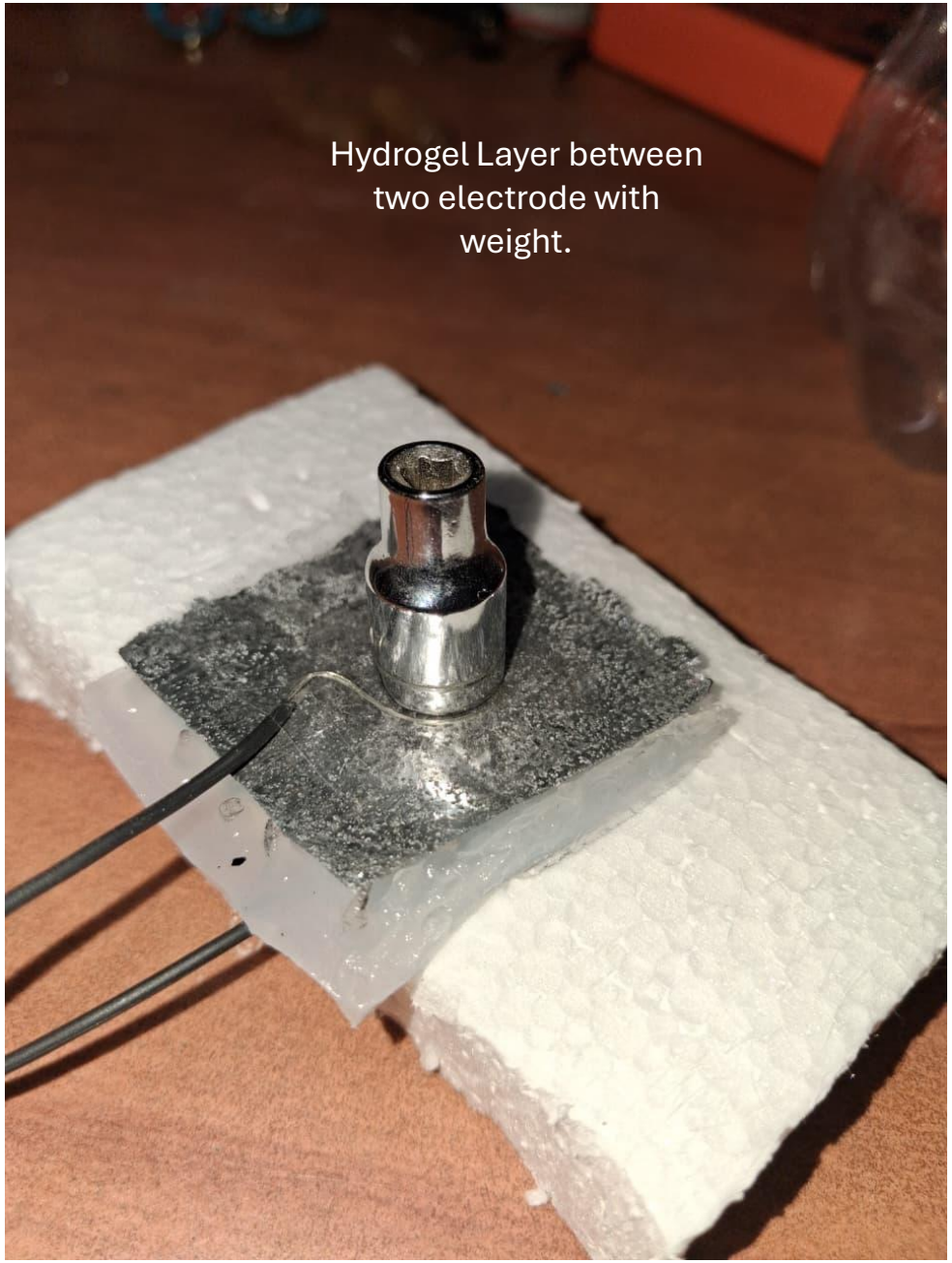
Making The gel conductive with Electrolyte immersion

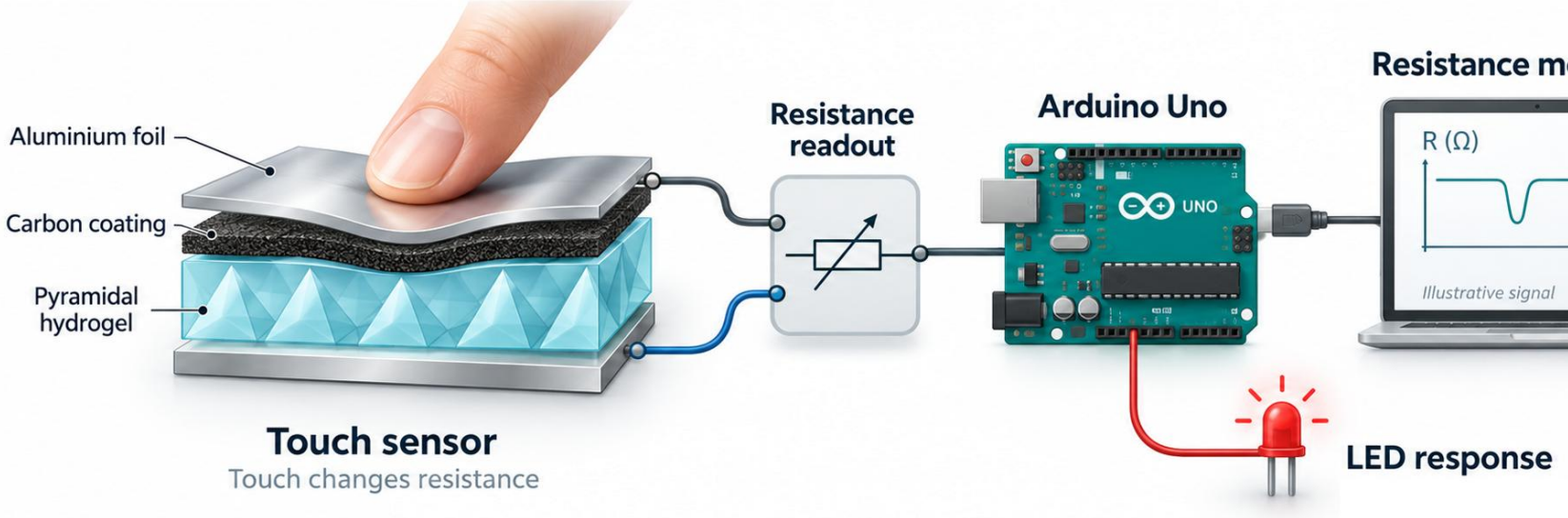




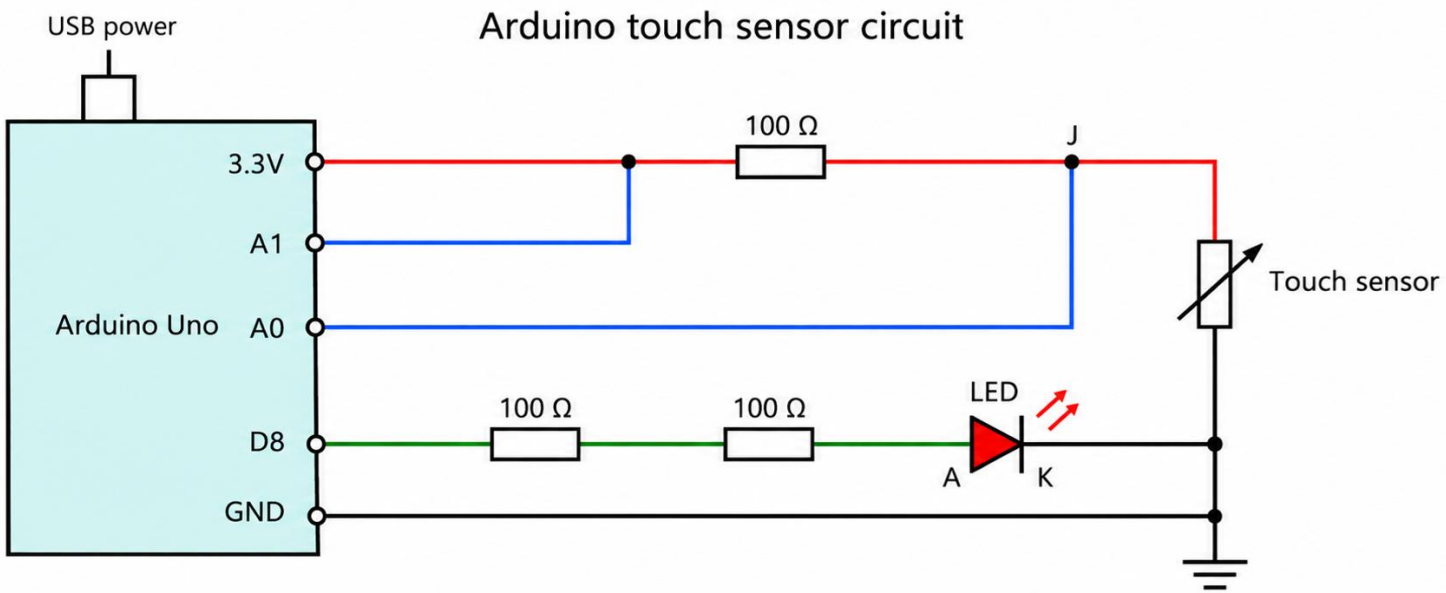

Hydrogel Layer with Pyramidal Needles after 3 cycles of Freezing & Thawing


Bottom Electrode, Al Sheet
Top Electrode, Al Sheet Coated with Activated Carbon Paint


Hydrogel Layer between two electrode with weight.







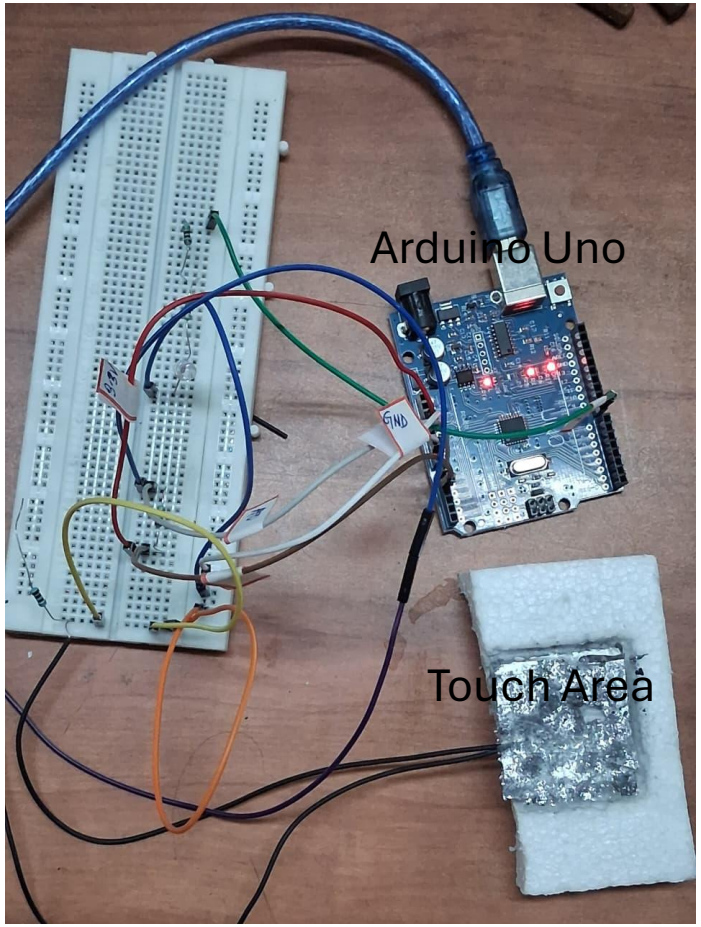


Fig. Circuit Design, Uno Integration

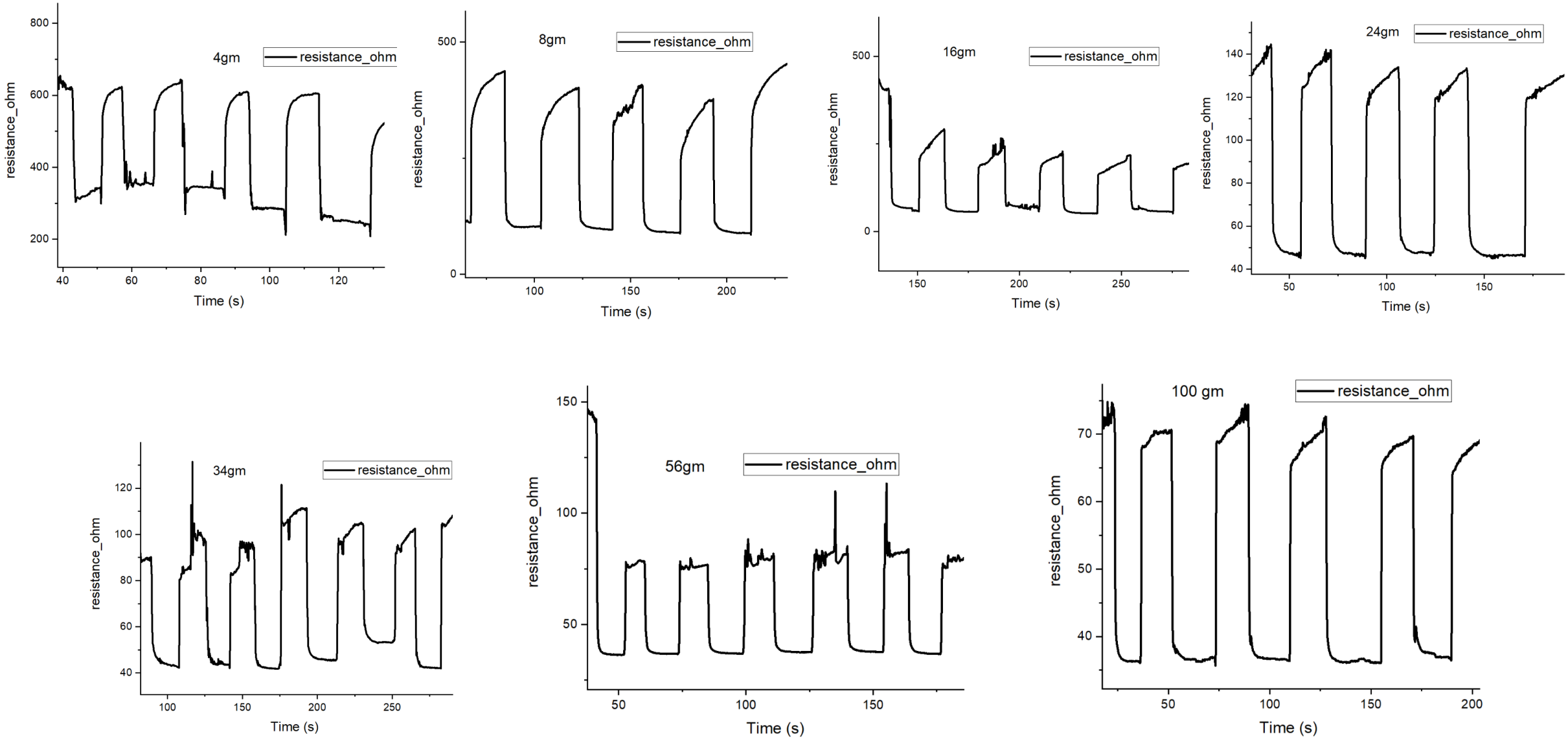
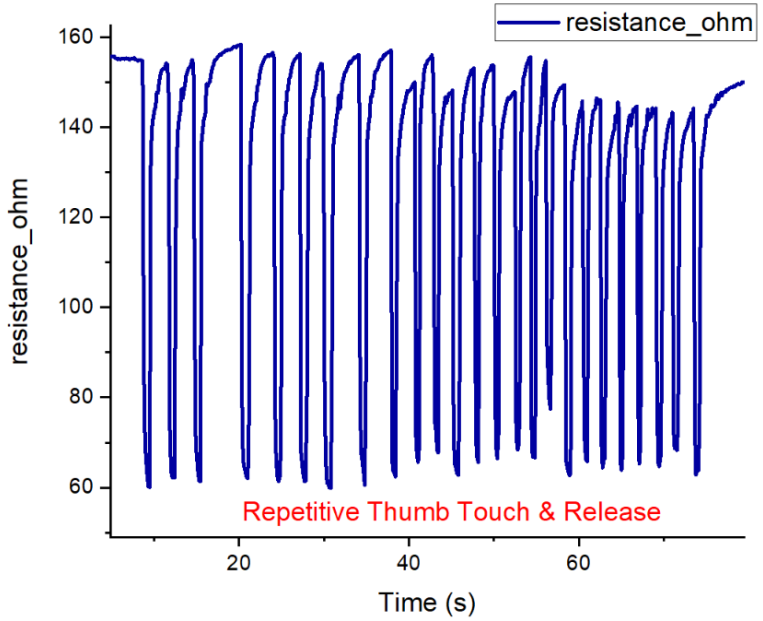
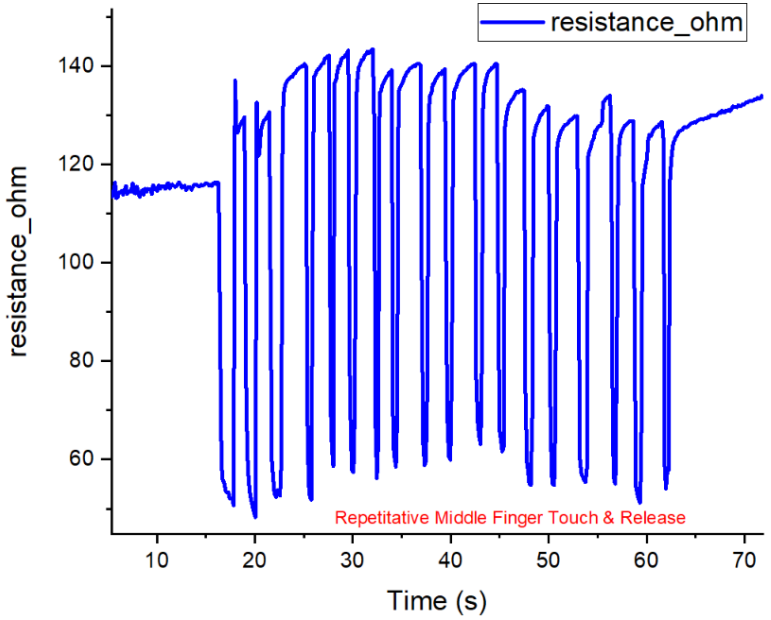
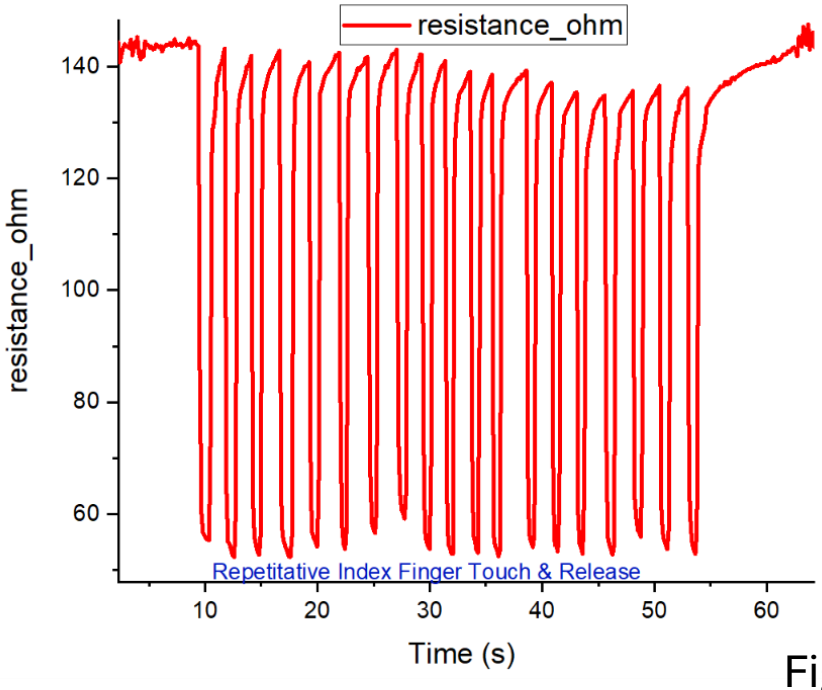
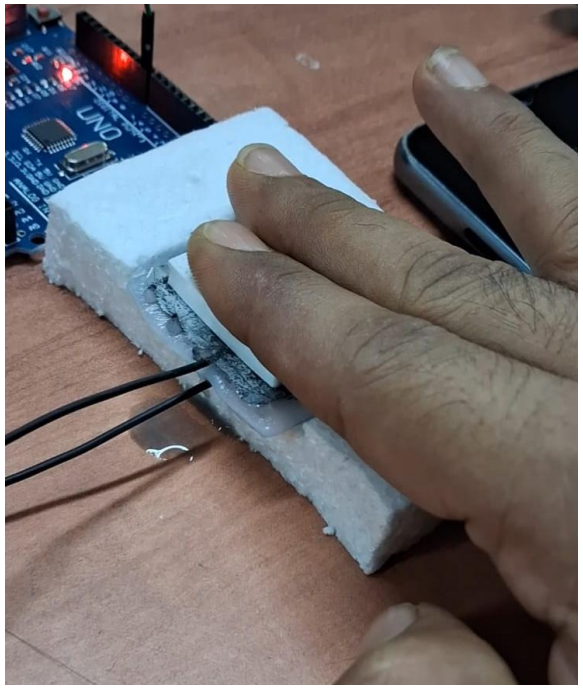


Fig. : Changes in resistance under various weight from 4gm to 100 gm. As the Pyramidal structures have some thawing problem hence the baseline is mostly varying. (Needs to be optimized after several rounds of freezing and thawing.)









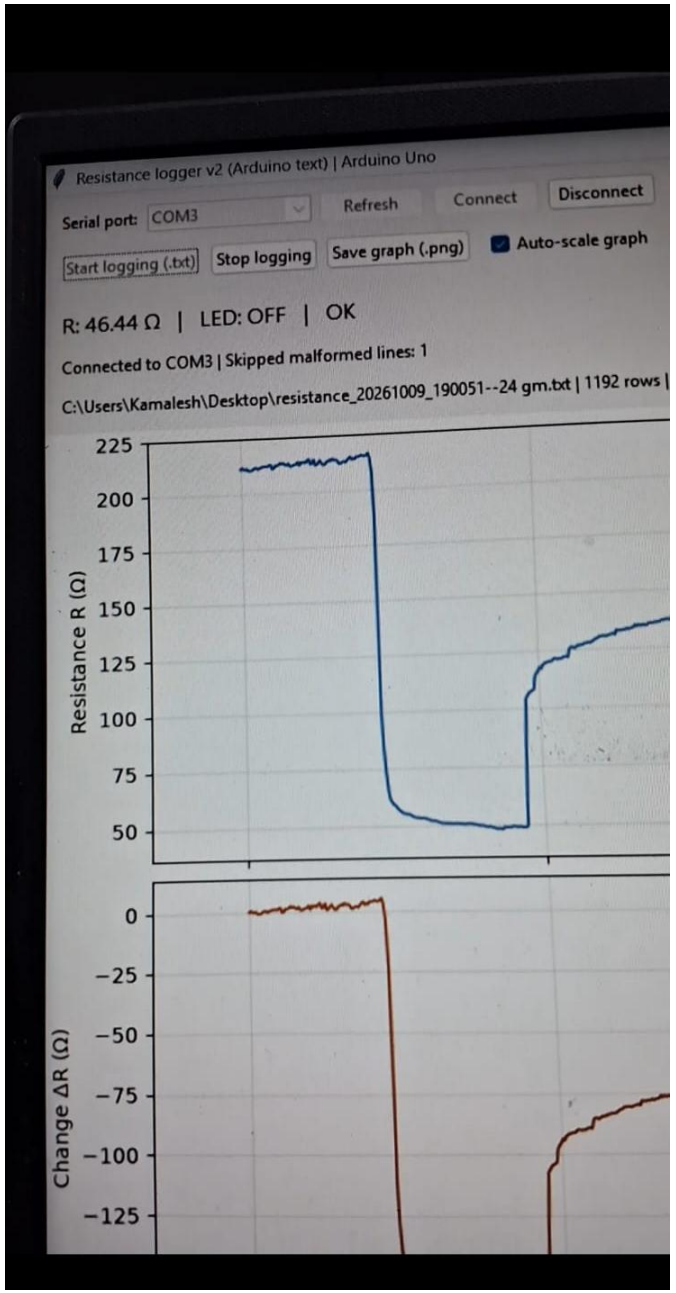
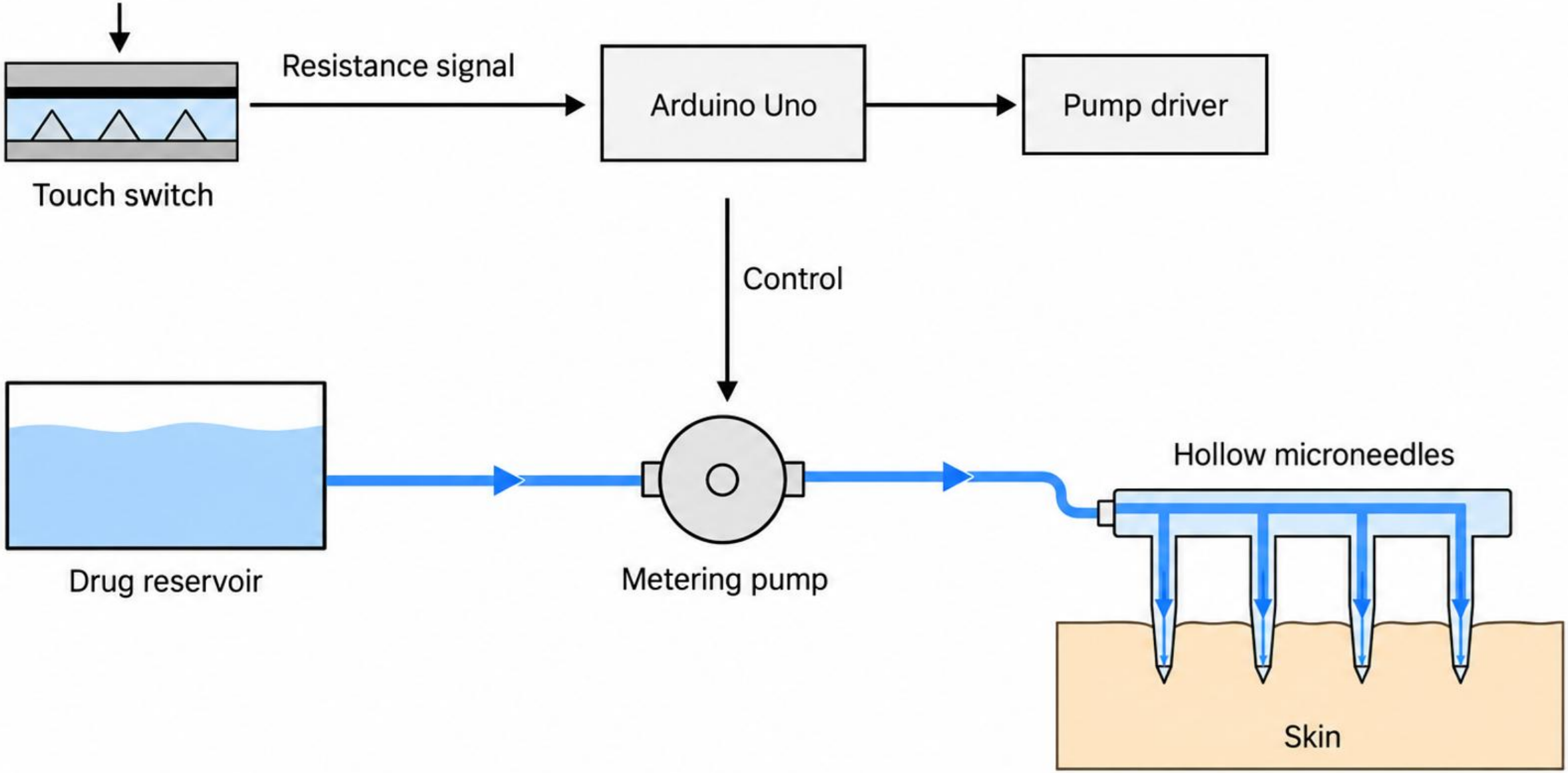
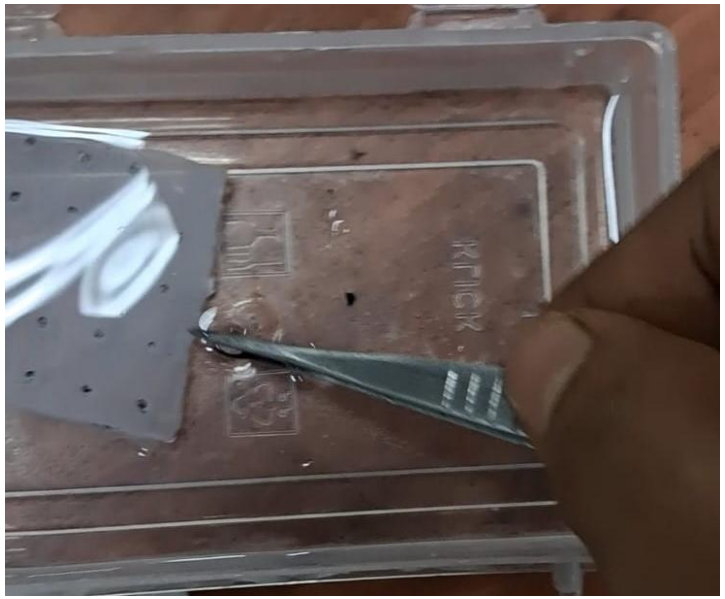
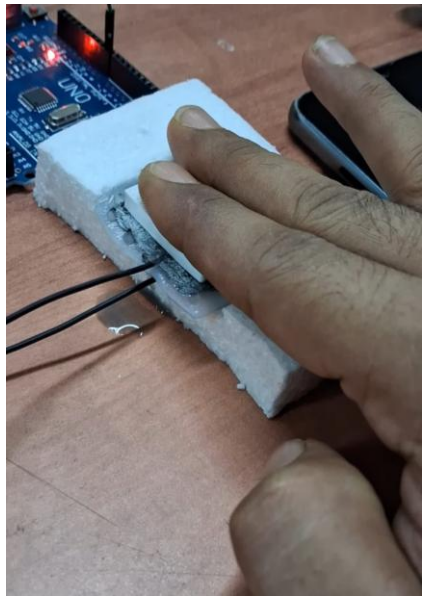


Fig. : changes in Resistance under various finger touch and release.

Potential Utility : Touch Controlled Precision Dosing of Drug





Various Images.